1(a) His explanation is not correct as even though the force that the truck acts on him is equal and opposite to the force that he acts on the truck, the 2 forces act on different bodies [APPLICATION] so they do not cancel out. [CONCLUSION]

As Adam pushes the truck, the truck exerts a friction on the surface of the roadside [be specific: friction at where?]. By Newton's 3rd law [CONCEPT], the surface of the roadside will exert an equal and opposite friction back on the truck. [APPLICATION]

The truck does not move because the friction that the surface of the roadside exerted on the truck and the force exerted on the truck by Adam are acting on the same object and are equal and opposite, therefore there is no net force on the truck, [APPLICATION] and hence by Newton's 1st law [CONCEPT], it doesn't move [CONCLUSION].

(b)(i) The principle of conservation of momentum states that total momentum of the system before the collision is equal to the total momentum of the system after the collision, provided no net external force acts on the system.

(ii) 1. Total initial momentum of the system is 0 since nobody is moving.

By conservation of momentum [CONCEPT],
Taking upwards as positive,
Total initial momentum of system = total final momentum of system

$$0 = M_{balloon}V_{balloon} + m_{man}v_{man}$$
$$0 = (320)V_{balloon} + 80(2.5)$$
$$V_{balloon} = -0.625 \text{ m s}^{-1}$$

Since upwards is taken to be positive, -0.625 m s^{-1} would indicate that the balloon is moving in the negative direction, which is downwards.

2. Speed of the balloon is zero [because initial total momentum = 0]